

F20DX COURSEWORK

Clark, Niamh

Table of Contents

<i>Introduction</i>	2
<i>Dataset Description</i>	3
<i>Data Exploration</i>	4
<i>Data Preparation</i>	6
<i>Decision Tree Experiments</i>	7
<i>Logistic Regression Experiments</i>	11
<i>Comparison & Discussion</i>	14
<i>Conclusion</i>	16
<i>References</i>	17

Introduction

Predictive maintenance is an important application of machine learning within modern industrial systems. Traditional maintenance strategies, such as reactive maintenance and scheduled maintenance, can result in unnecessary operational costs, unexpected downtime, or equipment failures. Predictive maintenance aims to address these issues by using operational data and sensor measurements to identify potential failures before they occur.

Machine learning techniques are increasingly used in predictive maintenance environments because they are capable of identifying patterns and relationships within large volumes of operational data. This allows organisations to improve maintenance scheduling, reduce downtime, and increase operational reliability.

This project investigates the use of supervised machine learning algorithms for predictive maintenance classification using a synthetic industrial machine dataset. Decision Tree and Logistic Regression classifiers were selected due to their differing approaches to classification and interpretability.

The project evaluates the models using multiple performance metrics including accuracy, precision, recall, F1-score, confusion matrices, and ROC curve analysis. Additional experiments investigate overfitting behaviour, class imbalance mitigation, and operational tradeoffs associated with failure prediction systems.

The overall aim of the project is to determine which machine learning approach is most suitable for predictive maintenance classification while considering both predictive performance and practical industrial implications.

Dataset Description

The dataset used in this project was the AI4I 2020 Predictive Maintenance Dataset obtained from the UCI Machine Learning Repository. The dataset is synthetic but was designed to realistically represent industrial predictive maintenance scenarios.

The dataset contains 10,000 observations with operational sensor measurements and machine condition information. Features include air temperature, process temperature, rotational speed, torque, tool wear, and machine type classification.

The target variable indicates whether a machine failure occurred during a specific operational observation. The dataset also contains a “Failure Type” feature describing the category of failure. However, this feature was removed during preprocessing because it directly revealed target information and would therefore introduce target leakage into the machine learning models.

The dataset was highly imbalanced, with failure observations representing approximately 3.4% of all records. This imbalance created additional challenges for classification and required careful evaluation beyond simple accuracy measurements.

Decision Trees were selected because they are capable of modelling non-linear relationships and complex feature interactions without requiring extensive feature assumptions. Logistic Regression was selected as a comparative linear baseline classifier due to its simplicity, interpretability, and widespread use in binary classification problems. Comparing both approaches allowed investigation of how linear and non-linear models behave when applied to highly imbalanced predictive maintenance data.

Data Exploration

Initial exploration of the dataset was performed to investigate feature distributions, class imbalance, and relationships between operational variables.

The dataset was found to be highly imbalanced, with machine failures representing only approximately 3.4% of all observations. Figure 1 shows the distribution of the target classes.

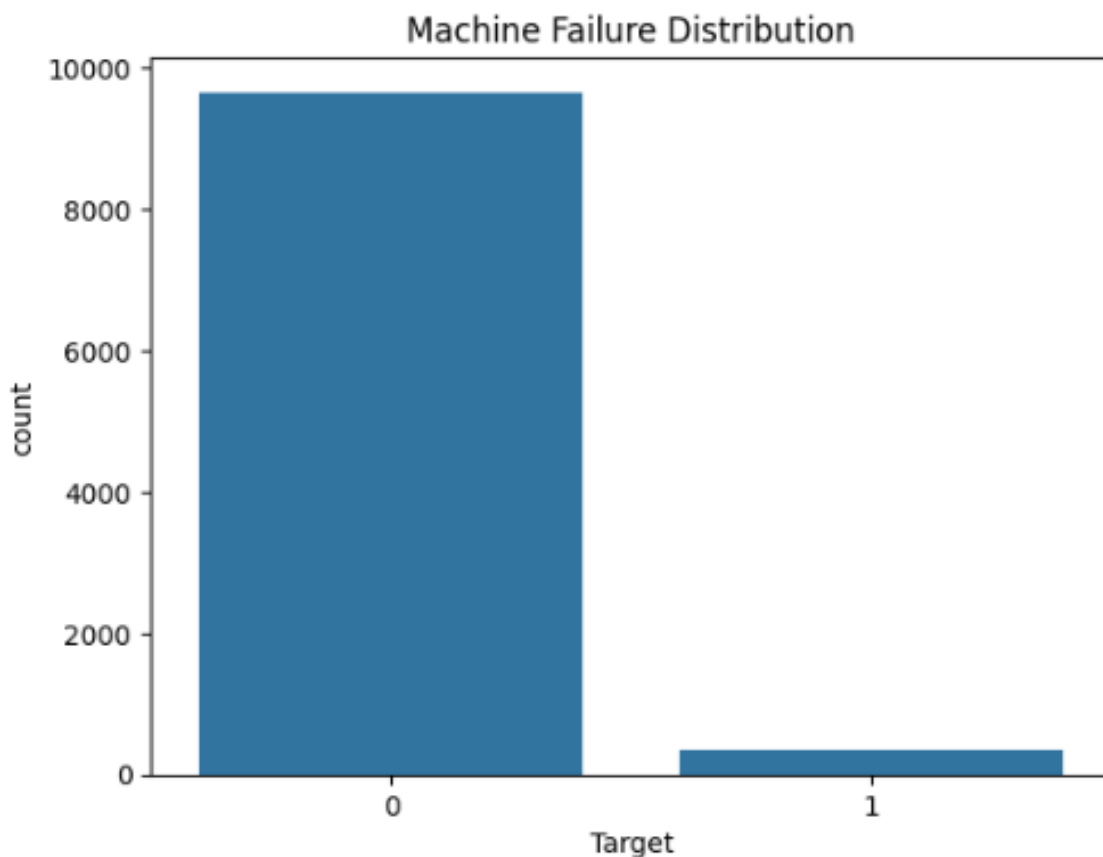


Figure 1 - Machine Failure Distribution

This imbalance presents a significant challenge for machine learning classification because a model may achieve high overall accuracy by predominantly predicting the majority non-failure class. As a result, additional evaluation metrics including precision, recall, F1-score, confusion matrices, and ROC curve analysis were considered necessary for meaningful model evaluation.

Correlation analysis was also performed to investigate relationships between numerical features within the dataset. Figure 2 shows the generated feature correlation heatmap.

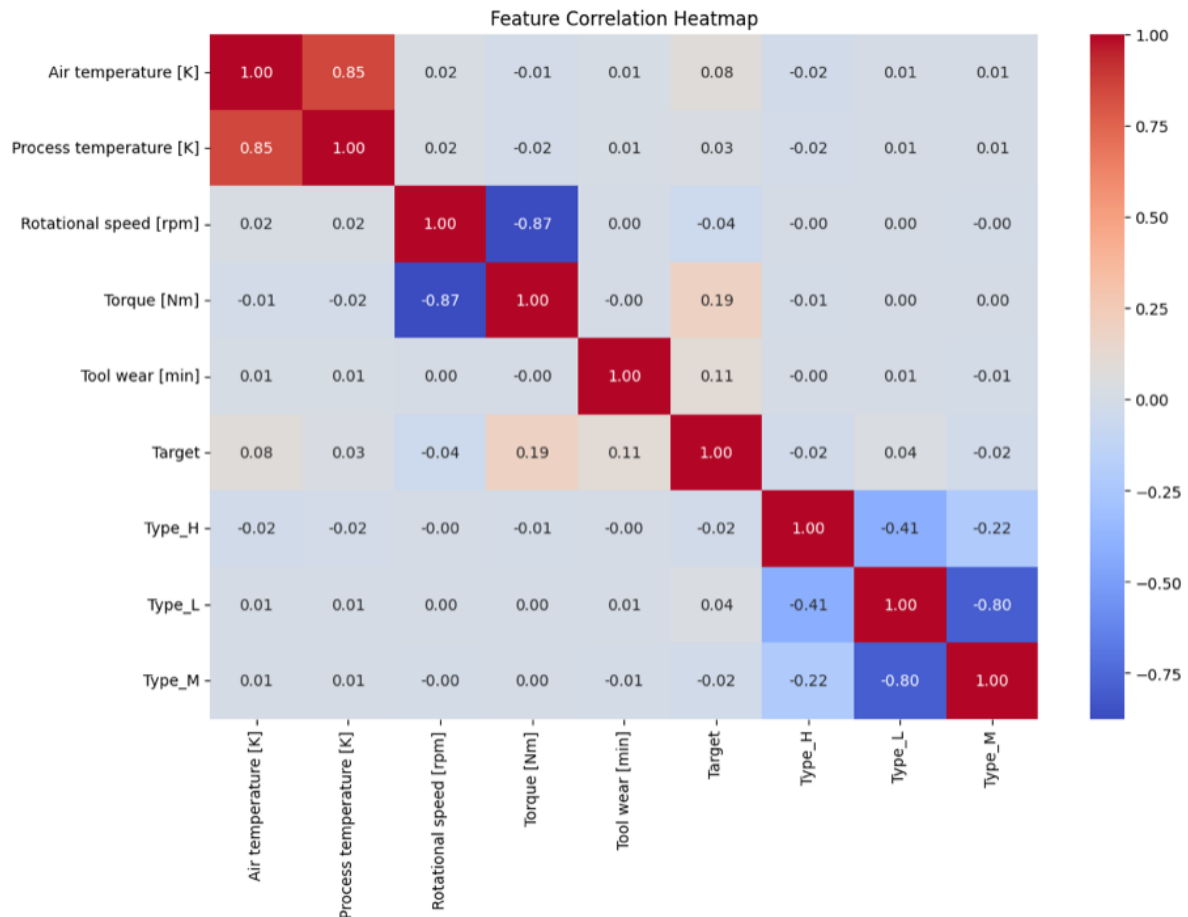


Figure 2 - Feature Correlation Heatmap

Several meaningful operational relationships were identified. Air temperature and process temperature exhibited a strong positive correlation, which is physically reasonable because process temperatures are influenced by surrounding environmental conditions.

Rotational speed and torque showed a strong negative correlation. This relationship is consistent with the inverse relationship often observed between rotational velocity and torque in approximately constant-power mechanical systems.

Correlations between individual features and the target variable were relatively weak, suggesting that machine failures are likely influenced by combinations of operational variables rather than a single dominant factor. This observation may help explain the relatively strong performance of the Decision Tree classifier later in the project, as tree-based models are capable of capturing non-linear feature interactions.

Data Preparation

Several preprocessing steps were applied to prepare the dataset for machine learning experiments.

Initially, the “UDI” and “Product ID” features were removed because they represented unique identifiers rather than operational characteristics. Including these features would not provide meaningful predictive information and could negatively influence model training.

The “Failure Type” feature was also removed from the dataset because it directly revealed information about the target classification. Including this feature would introduce target leakage, allowing machine learning models to indirectly access failure information during training and resulting in unrealistically optimistic performance.

Categorical machine type data was converted into numerical form using one-hot encoding. This transformation was necessary because machine learning algorithms require numerical inputs for training and prediction.

The dataset was then divided into training and testing subsets using an 80/20 split. Stratified sampling was used to preserve the original class distribution due to the significant imbalance between failure and non-failure observations.

Additional feature scaling was applied for the Logistic Regression experiments using standardisation. This ensured that features operated on similar numerical scales, which is important for distance and gradient-based optimisation methods used by Logistic Regression.

The scaler was fitted only on the training data before being applied to the testing data separately. This prevented data leakage between training and testing datasets.

Alternative imbalance mitigation techniques such as oversampling and synthetic minority generation (SMOTE) were considered; however, class weighting was selected due to its simplicity and direct integration within the Logistic Regression implementation.

These preprocessing steps ensured that the dataset was suitable for fair and reliable machine learning evaluation.

Decision Tree Experiments

Decision Tree classification was investigated as the primary non-linear machine learning approach for predictive maintenance prediction.

An initial baseline Decision Tree classifier was trained using default model parameters. The classifier achieved an overall accuracy of 97.6% on the testing dataset. However, due to the significant class imbalance present within the dataset, additional evaluation metrics were required to assess performance more meaningfully.

The baseline model achieved a recall score of 0.63 for the machine failure class, meaning that approximately 63% of actual failures were correctly identified. The confusion matrix showed that 25 machine failures were incorrectly classified as non-failures.

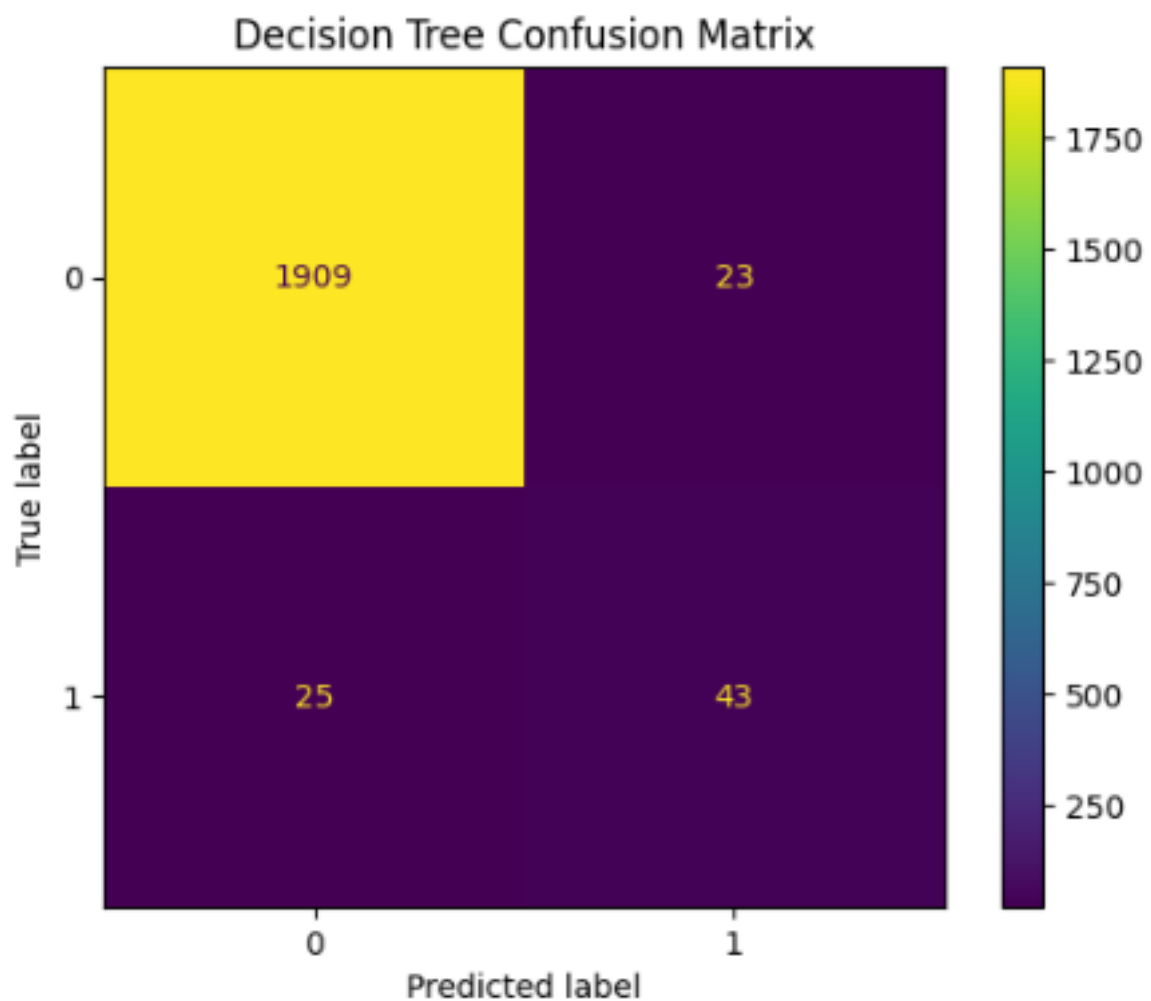


Figure 3 - Decision Tree Confusion Matrix

In predictive maintenance systems, false negatives are particularly important because undetected failures may result in equipment damage, operational downtime, or safety risks. Although the classifier achieved high overall accuracy, reducing missed failures remained an important consideration.

Additional experiments were performed to investigate the effect of Decision Tree depth on model performance and generalisation behaviour. Several maximum depth values were tested to evaluate potential underfitting and overfitting effects.



Figure 4 - Training vs Testing Accuracy for Different Tree Depths

The experiments demonstrated that increasing tree depth initially improved classifier performance. The model with a maximum depth of 10 achieved the strongest testing accuracy while maintaining strong recall performance.

However, the unrestricted Decision Tree achieved 100% training accuracy while producing lower testing accuracy than the depth-limited model. This behaviour provides strong evidence of overfitting, where the classifier memorised highly specific training patterns and noise that did not generalise effectively to unseen data.

Comparison between training and testing accuracy demonstrated the importance of controlling Decision Tree complexity to achieve an effective balance between learning capability and model generalisation.

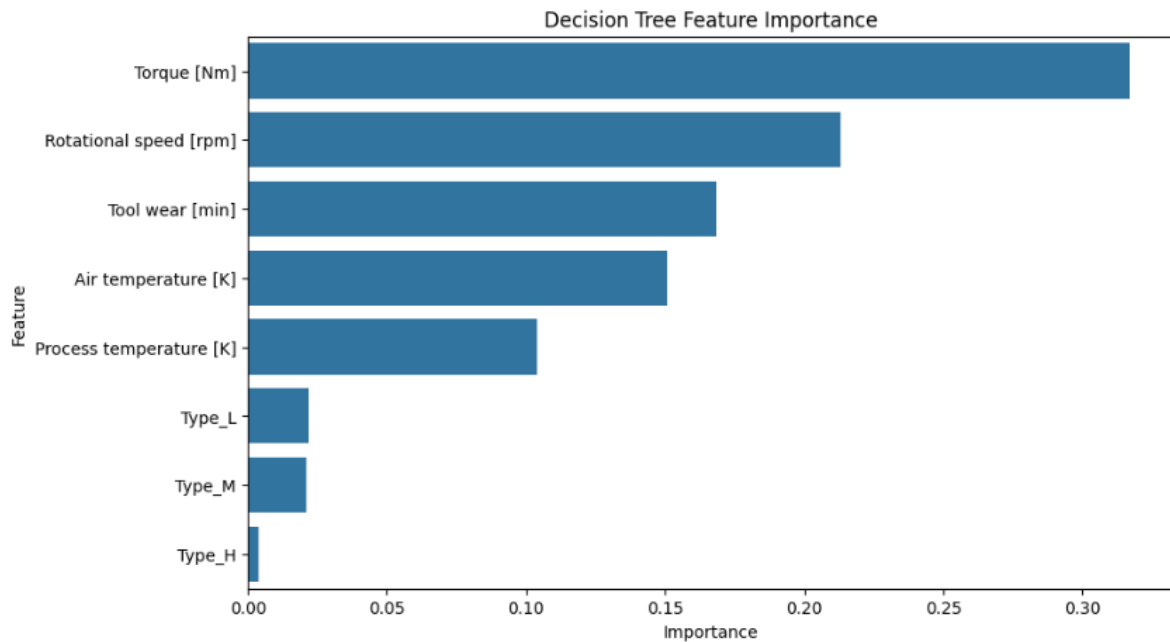


Figure 5 - Decision Tree Feature Importance

Feature importance analysis showed that torque, rotational speed, and tool wear were the most influential variables within the Decision Tree classifier. These findings are physically meaningful because mechanical loading, operational speed, and cumulative wear are realistic indicators of machine degradation and potential failure risk.

The Decision Tree classifier ultimately achieved the strongest balance between accuracy, recall performance, and operational practicality among the evaluated baseline models.

Additional Decision Tree parameters such as minimum sample splitting and pruning strategies could also be investigated to further reduce overfitting behaviour while maintaining strong recall performance.

Decision Tree Visualisation

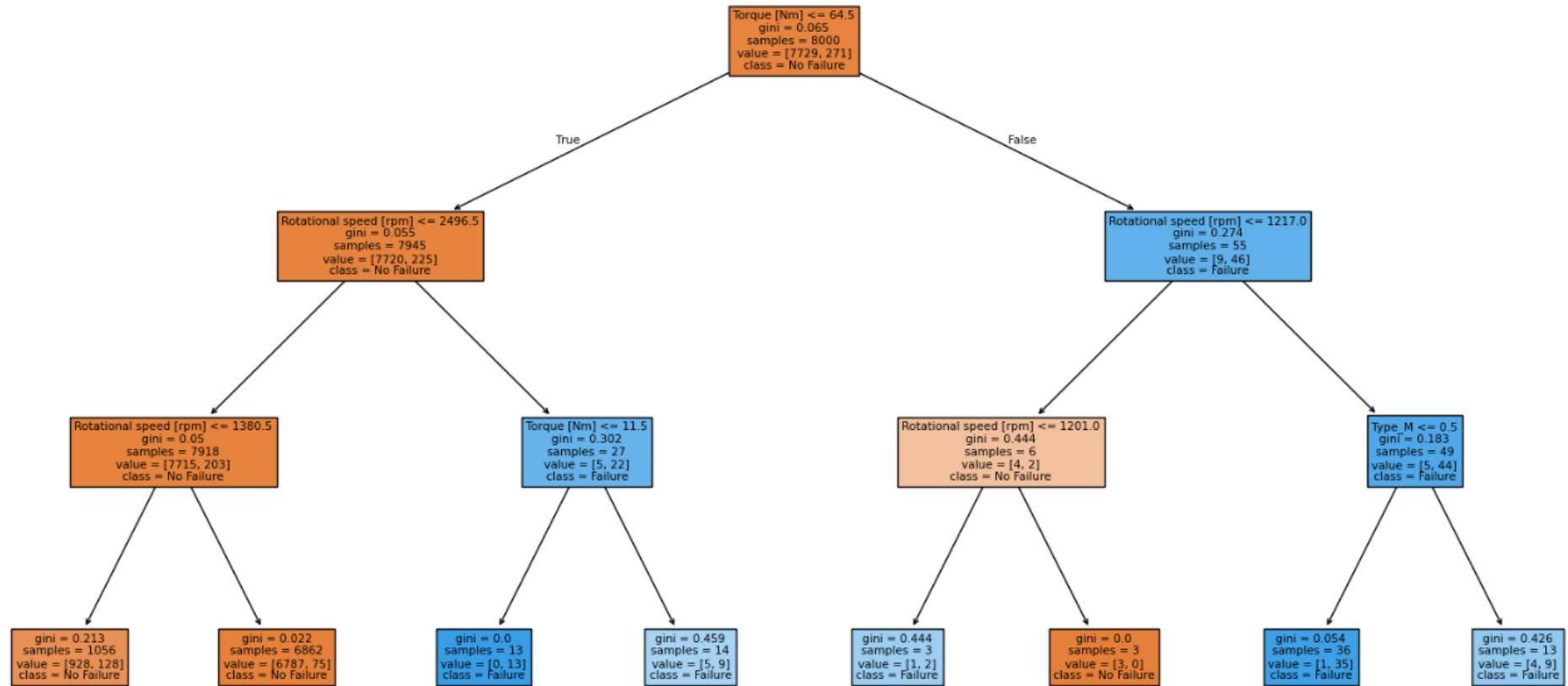


Figure 6 - Simplified Decision Tree Structure

Logistic Regression Experiments

Logistic Regression classification was investigated as a linear machine learning approach for predictive maintenance prediction.

Unlike Decision Trees, Logistic Regression is sensitive to differences in feature scale. Therefore, feature standardisation was applied during preprocessing to ensure that numerical variables operated on comparable scales prior to model training.

The baseline Logistic Regression classifier achieved an overall testing accuracy of 96.75%. However, despite this relatively high accuracy, the model performed poorly when identifying machine failures.

The classifier achieved a recall score of only 0.12 for the failure class, meaning that approximately 88% of machine failures were incorrectly classified as non-failures. The confusion matrix showed that 60 out of 68 failure observations were missed by the model.

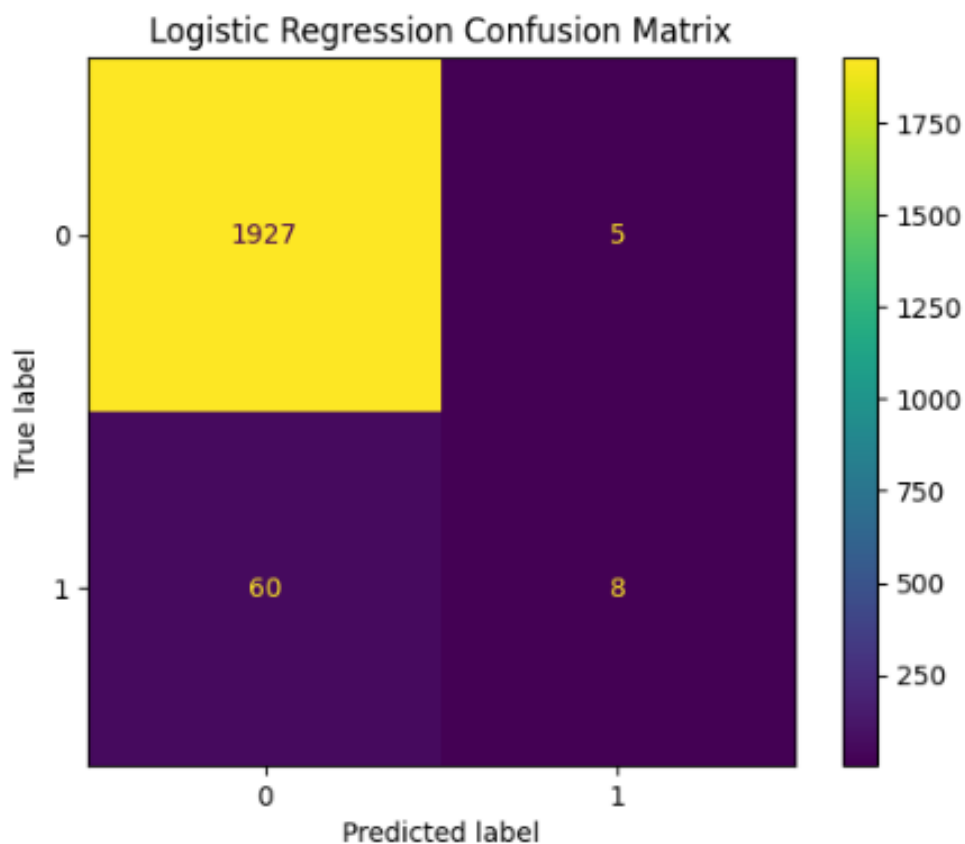


Figure 7 - Logistic Regression Confusion Matrix

These results demonstrate the limitations of relying solely on overall accuracy when evaluating highly imbalanced classification problems. Although the model achieved strong accuracy for the majority class, it struggled significantly to identify the minority failure class.

ROC curve analysis produced a substantially stronger evaluation of Logistic Regression performance. The baseline Logistic Regression model achieved an AUC score of approximately 0.90, indicating that the classifier was capable of separating failure and non-failure observations probabilistically despite poor fixed-threshold recall performance.

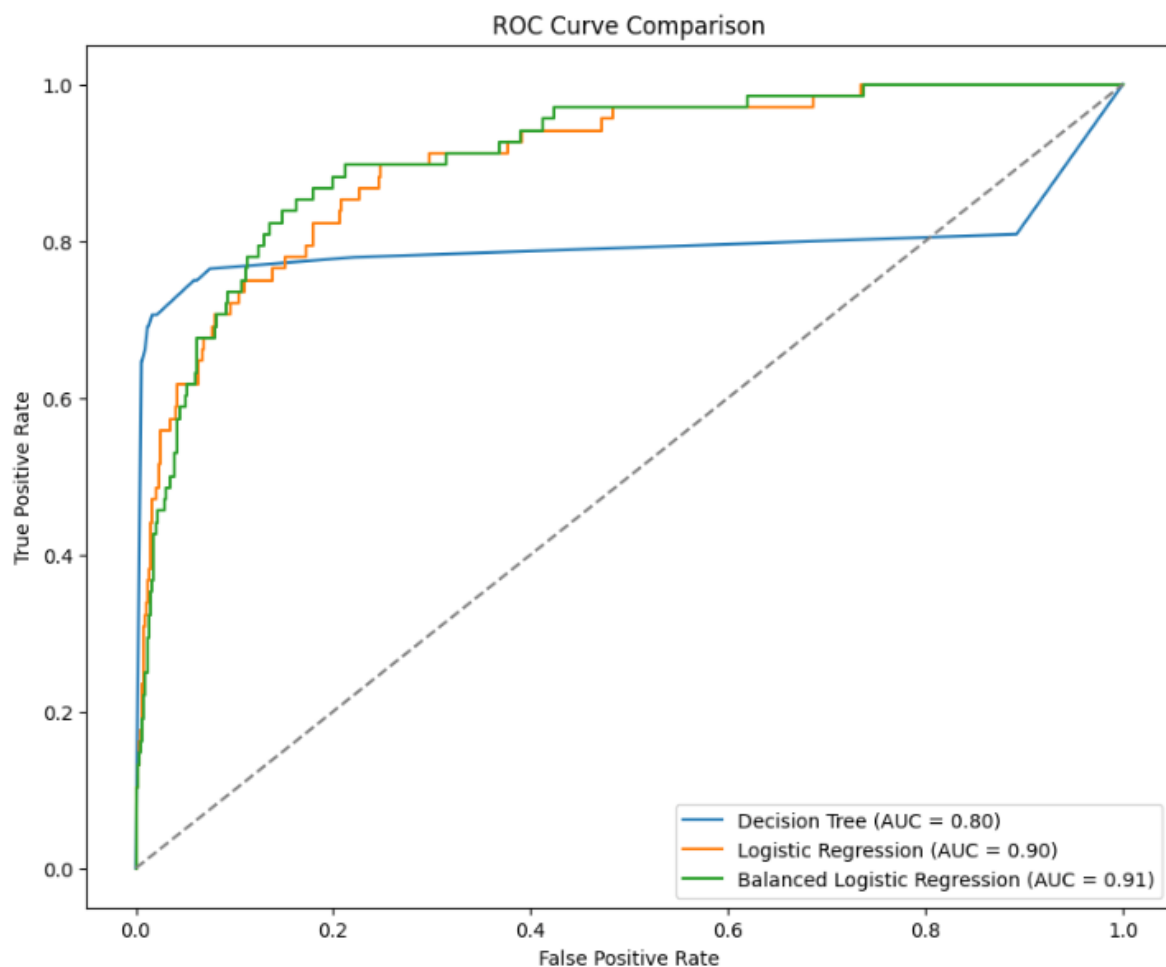


Figure 8 - ROC Curve Comparison for All Models

Additional experiments were therefore performed using balanced class weighting to investigate whether failure detection performance could be improved.

Applying balanced class weighting significantly improved recall performance for the failure class. The balanced Logistic Regression classifier achieved a recall score of 0.85, reducing missed machine failures from 60 cases to only 10.

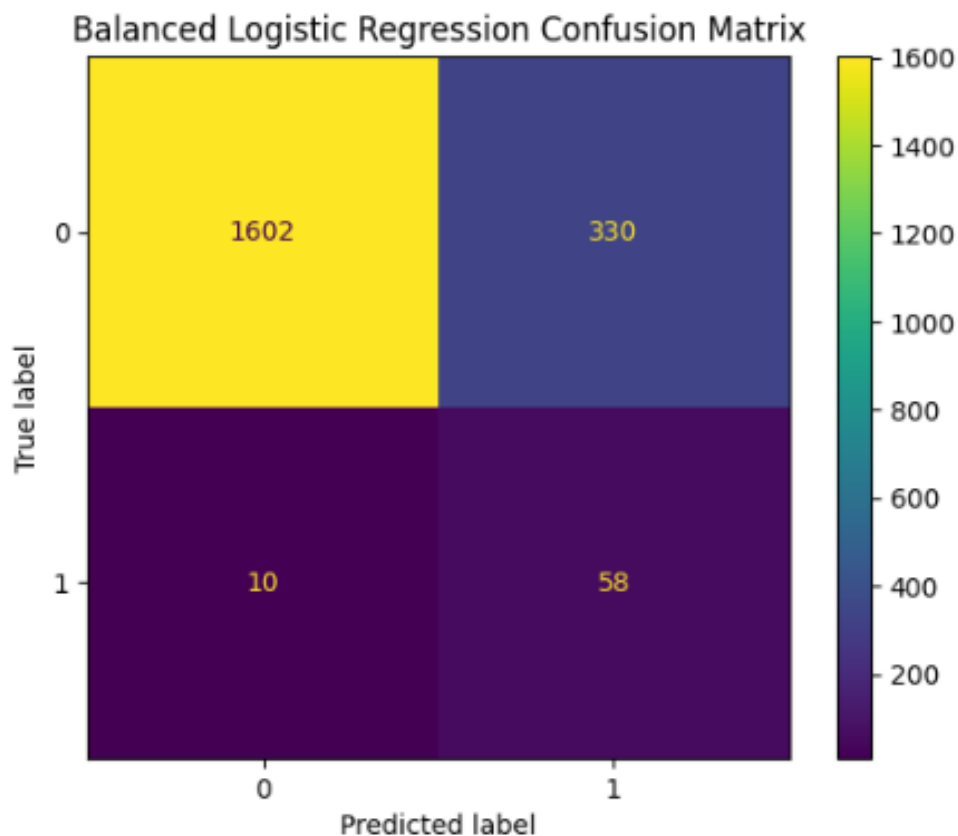


Figure 9 - Balanced Logistic Regression Confusion Matrix

However, this improvement introduced a substantial increase in false positives, with the classifier incorrectly identifying many healthy machines as failures. Overall testing accuracy also decreased to approximately 83%.

These experiments clearly demonstrated the precision-recall tradeoff commonly encountered in highly imbalanced classification problems. Increasing sensitivity to machine failures improved recall performance but also increased unnecessary maintenance predictions.

Despite weaker overall operational balance compared to the Decision Tree classifier, the Logistic Regression experiments provided valuable insight into the effects of class imbalance, threshold behaviour, and probability-based classification performance within predictive maintenance systems.

Comparison & Discussion

The experiments demonstrated clear differences between the machine learning approaches used for predictive maintenance classification.

The Decision Tree classifier achieved the strongest overall balance between accuracy, recall performance, and operational practicality. The model successfully identified approximately 63% of machine failures while maintaining strong overall classification accuracy and relatively low false positive rates.

The Logistic Regression classifier achieved similarly high overall accuracy but performed poorly when identifying machine failures at the default classification threshold. The classifier achieved a recall score of only 0.12 for the failure class, missing the majority of actual machine failures.

These results demonstrated the limitations of relying solely on overall accuracy when evaluating highly imbalanced datasets. Due to the significant imbalance between failure and non-failure observations, high accuracy could be achieved while still performing poorly on the minority failure class.

ROC curve analysis provided additional insight into classifier behaviour. Despite poor recall performance, the baseline Logistic Regression model achieved a strong AUC score, indicating that the classifier was capable of separating failure and non-failure observations probabilistically. This suggested that the default classification threshold was poorly suited to the imbalanced dataset rather than the model itself being fundamentally ineffective.

Applying balanced class weighting significantly improved Logistic Regression recall performance, increasing failure detection from 12% to 85%. However, this improvement introduced a substantial increase in false positives and reduced overall accuracy.

This demonstrated the precision-recall trade-off commonly encountered in predictive maintenance systems. Increasing sensitivity to machine failures reduced missed failures but also increased unnecessary maintenance predictions and operational intervention costs.

The Decision Tree classifier demonstrated stronger operational balance across the evaluated metrics. The model achieved substantially better recall performance than the baseline Logistic Regression classifier while avoiding the excessive false positive behaviour introduced by balanced class weighting.

The Decision Tree classifier also provided strong interpretability through feature importance analysis and tree structure visualisation. Torque, rotational speed, and tool wear were identified as the most influential operational variables, which aligned with realistic engineering expectations for mechanical degradation and failure behaviour.

Several limitations should also be considered. The dataset used in this project was synthetic and may not fully represent the complexity and variability of real industrial maintenance environments. Additionally, temporal operational behaviour and sequential sensor trends were not included within the experiments. Real predictive maintenance systems may therefore require more advanced feature engineering, time-series analysis, or ensemble learning techniques to achieve optimal real-world performance.

Overall, the experiments demonstrated that model evaluation for predictive maintenance systems should consider operational consequences alongside statistical performance metrics. In real industrial environments, the relative importance of recall, precision, and false positive rates would depend on the operational costs associated with missed failures and unnecessary maintenance actions.

Conclusion

This project investigated the use of supervised machine learning techniques for predictive maintenance classification using the AI4I 2020 Predictive Maintenance Dataset.

The experiments demonstrated that class imbalance had a major impact on classifier behaviour and evaluation. High overall accuracy alone was shown to be insufficient for meaningful evaluation because models could achieve strong accuracy while still performing poorly when identifying machine failures.

The Decision Tree classifier achieved the strongest overall balance between predictive performance and operational practicality. The model achieved high classification accuracy while also maintaining substantially stronger recall performance for machine failures compared to the baseline Logistic Regression classifier.

Additional experiments demonstrated the effects of model complexity, overfitting, and class imbalance mitigation. Increasing Decision Tree depth improved training performance but eventually reduced generalisation performance due to overfitting. Balanced class weighting significantly improved Logistic Regression recall performance but introduced a large increase in false positives.

Feature importance analysis showed that torque, rotational speed, and tool wear were the most influential variables for predicting machine failures. These findings aligned with realistic engineering expectations regarding operational stress, degradation, and mechanical wear.

Overall, the experiments demonstrated that effective predictive maintenance classification requires careful consideration of operational trade offs between recall, precision, and false positive behaviour. In real industrial environments, the optimal model configuration would depend on the operational consequences and costs associated with missed failures and unnecessary maintenance interventions.

Future work could investigate more advanced machine learning approaches such as Random Forests, Gradient Boosting, or XGBoost classifiers, as well as additional feature engineering and imbalance mitigation techniques to further improve predictive maintenance performance.

References

UCI (2020) UCI Machine Learning Repository. Available at: <https://archive.ics.uci.edu/dataset/601/ai4i+2020+predictive+maintenance+dataset> (Accessed: 14 May 2026).

Kaggle (2020) Machine Predictive Maintenance Classification Dataset. Available at: <https://www.kaggle.com/datasets/shivamb/machine-predictive-maintenance-classification/data> (Accessed: 14 May 2026).

Python Software Foundation (2026) Python Language Reference, version 3.13. Available at: <https://www.python.org> (Accessed: 14 May 2026).

Scikit-learn Developers (2026) Scikit-learn Documentation. Available at: <https://scikit-learn.org/stable/> (Accessed: 14 May 2026).